

ATTRIBUTE CLOSURE

Week 14



FUNCTIONAL DEPENDENCY AND ATTRIBUTE CLOSURE

Closure of an Attribute Set-

- The set of all those attributes which can be functionally determined from an attribute set is called as a closure of that attribute set.
- Closure of attribute set $\{X\}$ is denoted as $\{X\}^+$.



STEPS TO FIND CLOSURE OF AN ATTRIBUTE SET-

- Step-01:
- Add the attributes contained in the attribute set for which closure is being calculated to the result set.
- Step-02:
- Recursively add the attributes to the result set which can be functionally determined from the attributes already contained in the result set.



Example-

Consider a relation R (A , B , C , D , E , F , G) with the functional dependencies-

- $A \rightarrow BC$
- $BC \rightarrow DE$
- $D \rightarrow F$
- $CF \rightarrow G$



Closure of attribute A-

- $A^+ = \{ A \}$
- $= \{ A, B, C \}$ (Using $A \rightarrow BC$)
- $= \{ A, B, C, D, E \}$ (Using $BC \rightarrow DE$)
- $= \{ A, B, C, D, E, F \}$ (Using $D \rightarrow F$)
- $= \{ A, B, C, D, E, F, G \}$ (Using $CF \rightarrow G$)
- Thus,
- **$A^+ = \{ A, B, C, D, E, F, G \}$**



Closure of attribute D-

- $D^+ = \{ D \}$
- $= \{ D, F \}$ (Using $D \rightarrow F$)
- We can not determine any other attribute using attributes D and F contained in the result set.
- Thus,
- **$D^+ = \{ D, F \}$**



Closure of attribute set {B, C}-

- $\{ B , C \}^+ = \{ B , C \}$
- $= \{ B , C , D , E \}$ (Using $BC \rightarrow DE$)
- $= \{ B , C , D , E , F \}$ (Using $D \rightarrow F$)
- $= \{ B , C , D , E , F , G \}$ (Using $CF \rightarrow G$)
- Thus,
- $\{ \mathbf{B} , \mathbf{C} \}^+ = \{ \mathbf{B} , \mathbf{C} , \mathbf{D} , \mathbf{E} , \mathbf{F} , \mathbf{G} \}$
- Thus, attribute A is also a candidate key for that relation.



- Algorithm
- Let's see the algorithm to compute X^+
- Step 1 – $X^+ = X$
- Step 2 – repeat until X^+ does not change
 - For each FD $Y \rightarrow Z$ in F
 - If $Y \subseteq X^+$ then $X^+ = X^+ \cup Z$



Example 2

Consider a relation $R(A,B,C,D,E,F)$

F: $E \rightarrow A$, $E \rightarrow D$, $A \rightarrow C$, $A \rightarrow D$, $AE \rightarrow F$, $AG \rightarrow K$.

Find the closure of E or E^+

Solution

The closure of E or E^+ is as follows –

- $E^+ = E$
- $=EA$ {for $E \rightarrow A$ add A}
- $=EAD$ {for $E \rightarrow D$ add D}
- $=EADC$ {for $A \rightarrow C$ add C}
- $=EADC$ {for $A \rightarrow D$ D already added}
- $=EADCF$ {for $AE \rightarrow F$ add F}
- $=EADCF$ {for $AG \rightarrow K$ don't add k $AG \notin D^+$)}



Example 3

Let the relation $R(A,B,C,D,E,F)$

$F: B \rightarrow C, BC \rightarrow AD, D \rightarrow E, CF \rightarrow B$. Find the closure of B.

Solution

The closure for B is as follows –

- $B^+ = \{B, C, A, D, E\}$
- Closure is used to find the candidate keys of R and compute F^+
- Candidate key of R: X is a candidate keys of R if $X \rightarrow \{R\}$
- For example,
- $R(A,B,C,D,E,F)$ WHERE $F: A \rightarrow BC, B \rightarrow D, C \rightarrow DE, BC \rightarrow F$. Then, find the candidate keys of R.



Example 3 conti

Find the candidate keys of R.

Solution

- $A^+ = \{A, B, C, D, E, F\} = \{R\} \Rightarrow A$ is a candidate key
- $B^+ = \{B, D\} \Rightarrow B$ is not a candidate key
- $C^+ = \{C, D, E\} \Rightarrow C$ is not a candidate key
- $BC^+ = \{B, C, D, E, F\} \Rightarrow BC$ is not a candidate key
- Closure of F (F^+): F^+ is the set of all FDs that can be inferred/ derived from F. Using Armstrong Axioms repeatedly on F, we can compute all the FDs.



- Consider the relation scheme $R = \{E, F, G, H, I, J, K, L, M, N\}$ and the set of functional dependencies $\{\{E, F\} \rightarrow \{G\}, \{F\} \rightarrow \{I, J\}, \{E, H\} \rightarrow \{K, L\}, K \rightarrow \{M\}, L \rightarrow \{N\}$ on R . What is the key for R ?

- A. $\{E, F\}$
- B. $\{E, F, H\}$
- C. $\{E, F, H, K, L\}$
- D. $\{E\}$

Solution:

- Finding attribute closure of all given options, we get:
- $\{E, F\}^+ = \{EFGIJ\}$
- $\{E, F, H\}^+ = \{EFHGIJKLMN\}$
- $\{E, F, H, K, L\}^+ = \{EFHGIJKLMN\}$
- $\{E\}^+ = \{E\}$
- $\{EFH\}^+$ and $\{EFHKL\}^+$ results in set of all attributes, but EFH is minimal.
- So it will be candidate key. So correct option is (B).



- **Consider a relation scheme $R = (A, B, C, D, E, H)$ on which the following functional dependencies hold: $\{A \rightarrow B, BC \rightarrow D, E \rightarrow C, D \rightarrow A\}$. What are the candidate keys of R ?**
- a) AE, BE
- (b) AE, BE, DE
- (c) AEH, BEH, BCH
- (d) AEH, BEH, DEH

Solution:

- $(AE)^+ = \{ABECD\}$ which is not set of all attributes. So AE is not a candidate key. Hence option A and B are wrong.
- $(AEH)^+ = \{ABCDEH\}$
- $(BEH)^+ = \{BEHCDA\}$
- $(BCH)^+ = \{BCHDA\}$ which is not set of all attributes. So BCH is not a candidate key. Hence option C is wrong.
- So correct answer is D.

